EXPERIMENT: 34 IMPLEMENTING THE SIMULATION OF ERROR CORRECTION CODE – CRC IN JAVA/C

```
#include <stdio.h>
#include <string.h>

int main()
{
    char data[100], key[50], temp[150];
    int i, j, n, keyLen;

    printf("CRC ERROR DETECTION\n");
    printf("-------------------\n");

    printf("Enter data bits: ");
    scanf("%s", data);

    printf("Enter generator polynomial: ");
    scanf("%s", key);

    n = strlen(data);
    keyLen = strlen(key);

    /* Append zeros */
    strcpy(temp, data);

    for(i = 0; i < keyLen - 1; i++)
```

```
    strcat(temp, "0");

printf("\nData after adding zeros: %s\n", temp);

/* CRC calculation */
for(i = 0; i <= strlen(temp) - keyLen; i++)
{
    if(temp[i] == '1')
    {
        for(j = 0; j < keyLen; j++)
        {
            if(temp[i + j] == key[j])
                temp[i + j] = '0';
            else
                temp[i + j] = '1';
        }
    }
}

printf("CRC: ");

for(i = n; i < n + keyLen - 1; i++)
    printf("%c", temp[i]);

printf("\n");
```

```
    printf("Transmitted Data: %s", data);

    for(i = n; i < n + keyLen - 1; i++)
        printf("%c", temp[i]);

    printf("\n");

    return 0;
}
```

```
CRC ERROR DETECTION
-------------------
Enter data bits: 1101011011
Enter generator polynomial: 10011

Data after adding zeros: 11010110110000
CRC: 1110
Transmitted Data: 11010110111110


...Program finished with exit code 0
Press ENTER to exit console.
```